

ElectroMacroDiff V5.0

The 5-Day "Hackathon" Upgradation Plan & Architecture

Executive Summary

This document outlines the strategic pivot for ElectroMacroDiff V5.0 from a resource-intensive 30-day academic pipeline into a highly optimized, lab-ready 5-day "Hackathon" sprint. To circumvent the strict 15 GB VRAM limits of Google Colab T4 GPUs and avoid expensive compute clustering, we are deploying a decentralized 11-account Google Colab array. We trade massive volume for surgical speed using Parameter-Efficient Fine-Tuning (4-bit QLoRA), 2026 foundation models, and an FDA-style Target Product Profile (TPP) commercial deliverable.

1. What Are We Upgrading? (Old Plan vs. New Architecture)

To dominate the computational chemistry landscape on zero budget, we are executing a razor-sharp, resource-efficient strategy. The previous baseline architecture required ~450 A100 GPU hours for full joint training. The upgraded architecture is completely decoupled and rigorously optimized.

Component	Previous Baseline (Academic)	Upgraded V5.0 Architecture (Industry/Hackathon)
Hardware Strategy	Single Google Colab / Multi-GPU Joint Training	Decentralized 11-Account Colab Cluster. Bypasses Colab Pro requirements. Data synced via Google Cloud Bucket / Hugging Face.
Dataset Volume	5,000 to 10,000 pairs (ChEMBL + CrossDocked)	The Elite 500. Filtered strictly for the 500 most potent, high-quality macrocyclic JAK2 binders to accelerate convergence.
Generative Core (M5)	Standard DDIM Diffusion	PropMolFlow. A geometry-complete SE(3) flow matching framework fine-tuned via 4-bit QLoRA strictly on Transannular Attention layers.
Site Predictor (M3)	Transformer (MacTransformer)	GCN-Mamba. An O(n) architecture with a Selective State Space Transition Layer (S3TL). <i>Completely frozen</i> for the 5-day sprint to save memory and time.

Component	Previous Baseline (Academic)	Upgraded V5.0 Architecture (Industry/Hackathon)
Validation (M8)	AutoDock Vina / GNINA	Boltz-2 (Pairformer) & DiffDock. Operated strictly as zero-shot inference engines (wrapped in <code>torch.no_grad()</code>) for near-FEP accuracy without consuming optimizer VRAM.
Final Output	Ranked CSV/SDF with generated coordinates	Target Product Profile (TPP). An automated, FDA-style preclinical briefing document containing 3-5 hyper-optimized, synthesizable inhibitors negatively conditioned against SLC19A3 toxicity.

2. The Decentralized 11-Account Master Flow

We are decoupling the 11 modules of EMD V5.0 across an 11-account free-tier array to bypass single-user memory limits and frequent Colab disconnects.

Phase 1: Decentralized Pre-Computation (Accounts 1-3)

- **Domain Owner:** Sriram and Team.
- **Objective:** Offload static feature extraction so the generative core's memory is reserved entirely for the neural network.
- **Execution:** Use M1 (ElectroEncoder) and M2 (Pocket Encoder) to parse the JAK2 PDB structures. Extract explicit 3D electron density features (Gasteiger charges, Wiberg bond orders, and ESP features) and build a static PointNet++ graph.
- **Data Hand-off:** The resulting tensors are instantly pushed to a central Google Cloud Bucket.

Phase 2: Extreme QLoRA Generative Core (Accounts 4-6)

- **Domain Owner:** Main Generation Team.
- **Objective:** Train the heavy **SE(3)** flow matching generative framework within a 15 GB VRAM limit.
- **Execution:** The M3 GCN-Mamba model is loaded as a completely frozen asset to predict cyclization attachment points. The M5 Generative Core (PropMolFlow) is loaded in 4-bit normal float. We inject 16-bit brain float LoRA matrices specifically into the Transannular Attention layers.
- **Strategy:** Run heavily micro-batched training loops (2-4 molecules at a time) with gradient accumulation over the Elite 500 dataset to output 500-1,000 generated macrocycles in 48 hours.

Phase 3: The Zero-Shot Validation Gauntlet (Accounts 7-9)

- **Domain Owner:** Rahul and Team.
- **Objective:** Fast, mathematically rigorous filtering without training overhead.
- **Execution:** Download the generated 3D molecules. Run them through Boltz-2 and DiffDock. Because these are wrapped in `torch.no_grad()`, they act as pure zero-shot inference engines achieving near-Free-Energy Perturbation (FEP) accuracy.
- **Filtration:** Molecules are ranked by binding affinity; everything is aggressively cut except the top 10 candidates.

Phase 4: Synthesizability & Commercial Delivery (Accounts 10-11)

- **Objective:** De-risk the assets for pharma executives.
- **Agentic Retrosynthesis:** Push the top 10 candidates through AiZynthFinder. If the ring-closure chemistry is impossible, the molecule is programmatically rejected.
- **The "Killer Feature" (Safety):** Execute a final screening to confirm negative conditioning against SLC19A3, ensuring the drug mathematically avoids the Wernicke's encephalopathy toxicity liability.
- **Output:** The surviving 3 to 5 molecules are automatically compiled into the final Target Product Profile (TPP) document.

3. The 5-Day Sprint Timeline

- **Day 1: The Aggressive Data Cut.** Sriram's parallel instances extract M1/M2 static tensors for the Top 500 ChEMBL macrocyclic JAK2 binders. Data pushed to Cloud.
- **Days 2-3: Extreme QLoRA Generation.** Accounts 4-6 run heavily micro-batched 4-bit QLoRA on the PropMolFlow model using the pre-computed static tensors. Model outputs 500-1,000 generated structures.
- **Day 4: Zero-Shot Gauntlet.** Rahul's parallel instances run Boltz-2 and DiffDock inference. Bottom 99% of molecules are aggressively culled, leaving the Top 10.
- **Day 5: Synthesizability & TPP.** Retrosynthetic analysis via AiZynthFinder, SLC19A3 toxicity validation, and automated compilation of the FDA-style Target Product Profile.

4. Conclusion

By leveraging decentralized computing, static pre-computation, extreme parameter-efficient fine-tuning (4-bit QLoRA), and zero-shot foundation model inference, this upgraded EMD V5.0

architecture transforms a hardware-limited environment into an industry-grade drug discovery pipeline. It trades broad, generalized training for razor-sharp, target-specific speed, producing lab-ready, synthesizable macrocycles in a fraction of the traditional timeline.